Chapter 1: The Real and Complex Number Systems

Section: Appendix — Construction of the Real Numbers

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation and Overview

Step 1: Dedekind Cuts

Step 2: Ordering the Cuts

Step 3: The Least-Upper-Bound Property

Step 4: Addition of Cuts

Step 5: Order Compatibility with Addition

Steps 6–7: Multiplication

Steps 8–9: Embedding $\mathbb{Q}$ into $\mathbb{R}$

Summary

Motivation and Overview

Recall: Theorem 1.19

Theorem 1.19

There exists an ordered field $\mathbb{R}$ which has the least-upper-bound property. Moreover, $\mathbb{Q}$ is a subfield of $\mathbb{R}$.

This was *stated* earlier without proof. The entire appendix is devoted to proving this theorem.

Why This Appendix Matters

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- This appendix provides the explicit construction of $\mathbb{R}$ from $\mathbb{Q}$
- The method: Dedekind cuts — invented by Richard Dedekind in 1872

Construction Roadmap

Steps 1–2: Define cuts and order them $\longrightarrow$ $\mathbb{R}$ is an ordered set

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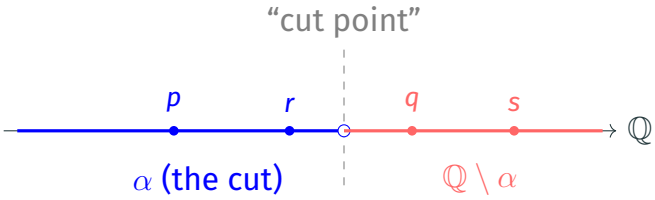
Step 1: Dedekind Cuts

The Big Idea: Cuts as Real Numbers



A cut α is a subset of $\mathbb{Q}$ that is "everything to the left" of some dividing point.

Step 1: Definition of a Cut

Definition: Cut (Dedekind Cut)

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(II) If $p \in \alpha$, $q \in \mathbb{Q}$, and $q < p$, then $q \in \alpha$

(III) If $p \in \alpha$, then $p < r$ for some $r \in \alpha$

The members of $\mathbb{R}$ will be the cuts. Letters $p, q, r, \ldots$ denote rationals; $\alpha, \beta, \gamma, \ldots$ denote cuts.

Consequences of the Definition

Property (II) immediately implies two useful facts:

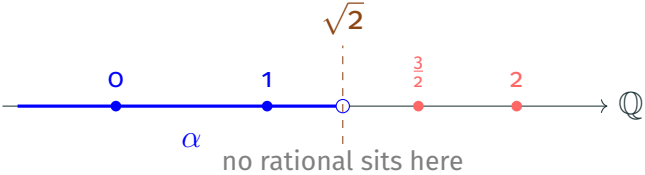
1. If $p \in \alpha$ and $q \notin \alpha$, then $p < q$
2. If $r \notin \alpha$ and $r < s$, then $s \notin \alpha$

Intuition: Elements of α are always *smaller* than non-elements.

Example: The Cut for $\sqrt{2}$

Example

Define $\alpha = \{p \in \mathbb{Q} : p < 0 \text{ or } p^2 < 2\}$.



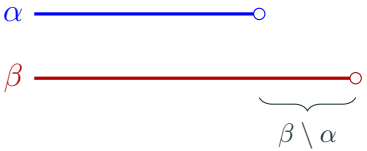
Step 2: Ordering the Cuts

Step 2: The Order Relation

Definition: Order on $\mathbb{R}$

For cuts α, β, define

$$\alpha < \beta \quad \Longleftrightarrow \quad \alpha \subsetneq \beta \quad (\alpha \text{ is a proper subset of } \beta).$$



$\alpha < \beta$

Proof: $\mathbb{R}$ Is an Ordered Set

Verification of Definition 1.5

Transitivity: If $\alpha < \beta$ and $\beta < \gamma$, then $\alpha \subsetneq \beta \subsetneq \gamma$, so $\alpha \subsetneq \gamma$.

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At most one holds: For any α, β, at most one of

$$\alpha < \beta, \qquad \alpha = \beta, \qquad \beta < \alpha$$

can hold.

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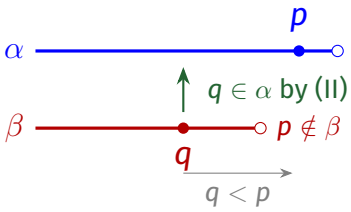
Trichotomy: At least one holds.

Proof of Trichotomy — Key Idea

Assume $\alpha \not\subset \beta$. Then some $p \in \alpha$ satisfies $p \notin \beta$:



Since $p \notin \beta$, *every* $q \in \beta$ satisfies $q < p$, so $q \in \alpha$.

Thus $\beta \subset \alpha$, and since $\beta \neq \alpha$: $\beta < \alpha$.

Proof of Trichotomy

At least one relation holds

Assume $\alpha < \beta$ and $\alpha = \beta$ both fail.

Then $\alpha \not\subset \beta$, so $\exists\, p \in \alpha$ with $p \notin \beta$.

Pick any $q \in \beta$. Since $p \notin \beta$, we get $q < p$ (by the consequence of (II)).

Since $q < p$ and $p \in \alpha$, property (II) gives $q \in \alpha$.

Thus every element of β belongs to α: $\beta \subset \alpha$.

Since $\beta \neq \alpha$, we conclude $\beta < \alpha$.

Step 3: The Least-Upper-Bound Property

Step 3: The Key Result

Theorem (Step 3)

The ordered set $\mathbb{R}$ has the least-upper-bound property.

Strategy: Given $A \subset \mathbb{R}$ nonempty and bounded above by β, define

$$\gamma = \bigcup_{\alpha \in A} \alpha$$

and show $\gamma = \sup A$.

Proof: γ Is a Cut

Verifying property (I): $\gamma \neq \emptyset$ and $\gamma \neq \mathbb{Q}$

- A is nonempty, so $\exists\, \alpha_0 \in A$. Since $\alpha_0 \neq \emptyset$ and $\alpha_0 \subset \gamma$, we get $\gamma \neq \emptyset$.
- $\gamma \subset \beta$ (since every $\alpha \in A$ satisfies $\alpha \subset \beta$), and $\beta \neq \mathbb{Q}$, so $\gamma \neq \mathbb{Q}$.

Proof: γ Is a Cut

Verifying property (II): if $p \in \gamma$ and $q < p$, then $q \in \gamma$

Pick $p \in \gamma$. Then $p \in \alpha_1$ for some $\alpha_1 \in A$.

If $q < p$, then $q \in \alpha_1$ (since α_1 is a cut), hence $q \in \gamma$.

Proof: γ Is a Cut

Verifying property (III): γ has no largest element

Pick $p \in \gamma$. Then $p \in \alpha_1$ for some $\alpha_1 \in A$.

Since α_1 has no largest element, $\exists\, r \in \alpha_1$ with $r > p$.

Then $r \in \gamma$ and $r > p$, so γ has no largest element.

$\boxed{\gamma \in R}$

Proof: $\gamma = \sup A$

γ is the least upper bound

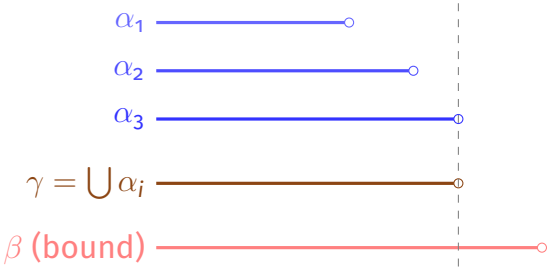
Upper bound: For every $\alpha \in A$, $\alpha \subset \gamma$ (by definition of union), so $\alpha \leq \gamma$.
Least: Suppose $\delta < \gamma$. Then $\exists\, s \in \gamma$ with $s \notin \delta$.

Since $s \in \gamma$, we have $s \in \alpha$ for some $\alpha \in A$.

Hence $\delta < \alpha$, so δ is *not* an upper bound of A.

$\boxed{\gamma = \sup A}$

Visual Summary: $\gamma = \sup A$



The union γ reaches exactly as far as the "largest" cut in A.

Step 4: Addition of Cuts

Step 4: Defining Addition

Definition: Addition of Cuts

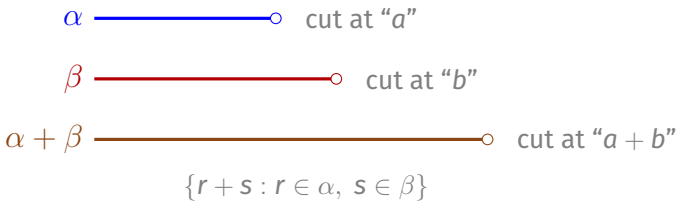
For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha + \beta = \{r + s : r \in \alpha,\ s \in \beta\}$$

Definition: The Zero Cut

Define $0^* = \{p \in \mathbb{Q} : p < 0\}$.

Visualizing Cut Addition



Adding cuts "adds their boundary points."

Axiom (A1): $\alpha + \beta$ Is a Cut

Verifying (I): $\alpha + \beta \neq \emptyset$ and $\alpha + \beta \neq \mathbb{Q}$

$\alpha + \beta$ is nonempty (pick $r \in \alpha$, $s \in \beta$; then $r + s \in \alpha + \beta$).

Take $r' \notin \alpha$, $s' \notin \beta$. For any $r \in \alpha$, $s \in \beta$:

$$r < r', \quad s < s' \quad \Longrightarrow \quad r + s < r' + s'$$

So $r' + s' \notin \alpha + \beta$, hence $\alpha + \beta \neq \mathbb{Q}$.

Axiom (A1): $\alpha + \beta$ Is a Cut

Verifying (II): if $p \in \alpha + \beta$ and $q < p$, then $q \in \alpha + \beta$

Pick $p \in \alpha + \beta$, so $p = r + s$ with $r \in \alpha$, $s \in \beta$.

If $q < p$, then $q - s < r$, so $q - s \in \alpha$ (by (II) for α).

Then $q = (q - s) + s \in \alpha + \beta$.

Axiom (A1): $\alpha + \beta$ Is a Cut

Verifying (III): $\alpha + \beta$ has no largest element

Pick $p = r + s \in \alpha + \beta$.

Choose $t \in \alpha$ with $t > r$ (since α has no largest element).

Then $t + s > r + s = p$ and $t + s \in \alpha + \beta$.

$\boxed{\alpha + \beta \text{ is a cut}}$

Axioms (A2)–(A4): Commutativity, Associativity, Identity

Quick verifications

(A2) Commutativity: $\alpha + \beta = \beta + \alpha$

Follows from $r + s = s + r$ in $\mathbb{Q}$.

(A3) Associativity: $(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma)$

Follows from associativity in $\mathbb{Q}$.

(A4) Additive identity: $\alpha + 0^* = \alpha$

$\subset$: If $r \in \alpha$, $s < 0$, then $r + s < r$, so $r + s \in \alpha$.

$\supset$: Pick $p \in \alpha$, choose $r \in \alpha$ with $r > p$. Then $p - r \in 0^*$ and $p = r + (p - r)$.

Axiom (A5): Additive Inverse — Definition

Definition of $-\alpha$

For $\alpha \in \mathbb{R}$, define β to be the set of all $p \in \mathbb{Q}$ such that:

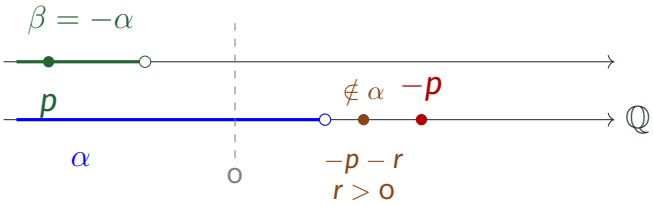
There exists $r > 0$ with $-p - r \notin \alpha$.

Equivalently: some rational number *strictly less than* $-p$ is not in α.

High-level idea: Obtain $-\alpha$ by flipping the "complement" of α (rationals $>$ boundary) across 0 (to get rationals $<$ -boundary). The buffer $r > 0$ ensures the boundary point itself is *excluded* from $-\alpha$.

Goal: Show $\beta \in \mathbb{R}$ and $\alpha + \beta = 0^*$.

Axiom (A5): Visualizing $-\alpha$



$p \in \beta$ because $-p - r \notin \alpha$: the point $-p$ lies safely beyond α, so p lies safely to the left of $-\alpha$.

Axiom (A5): $\beta = -\alpha$ Is a Cut

Verifying $\beta \in \mathbb{R}$

(I): If $s \notin \alpha$, set $p = -s - 1$. Then $-p - 1 = s \notin \alpha$, so $p \in \beta$. Also $q \in \alpha \Rightarrow -q \notin \beta$, so $\beta \neq \mathbb{Q}$.

(II): If $p \in \beta$ with witness $r > 0$, and $q < p$, then $-q - r > -p - r \notin \alpha$, so $q \in \beta$.

(III): Given $p \in \beta$ with witness r, set $t = p + r/2$. Then $t > p$ and $-t - r/2 = -p - r \notin \alpha$, so $t \in \beta$.

Axiom (A5): Proof that $\alpha + \beta = 0^*$

$\alpha + \beta \subset 0^*$

If $r \in \alpha$ and $s \in \beta$, then $-s \notin \alpha$, so $r < -s$, hence $r + s < 0$.

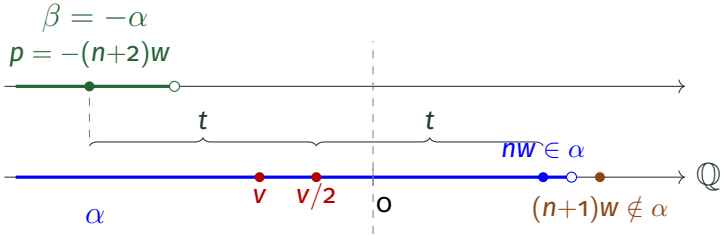
$0^* \subset \alpha + \beta$

High-level idea: Pick $v < 0$. Write $v = v/2 + v/2$ and split it as $v = (v/2 + t) + (v/2 - t)$, $t > 0$ chosen so that $v/2 + t \in \alpha$ and $v/2 - t \in \beta$.

Set $w = -v/2 > 0$. By the Archimedean property of $\mathbb{Q}$, $\exists\, n$ such that $nw \in \alpha$ but $(n+1)w \notin \alpha$.

Take $t = (n+1)w$. Then $v/2 + t = nw \in \alpha$, and $p := v/2 - t = -(n+2)w$ satisfies $-p - w = (n+1)w \notin \alpha$, so $p \in \beta$. Hence $v = nw + p \in \alpha + \beta$.

Axiom (A5): Visualizing $0^* \subset \alpha + \beta$



$v/2$ sits exactly halfway between p and nw: shifting by $t = (n+1)w$ to the right lands at $nw \in \alpha$, shifting by t to the left lands at $p \in \beta$.

Step 5: Order Compatibility with Addition

Step 5: Addition Preserves Order

Proposition (Step 5)

If $\alpha, \beta, \gamma \in \mathbb{R}$ and $\beta < \gamma$, then $\alpha + \beta < \alpha + \gamma$.

Proof

From the definition of $+$: $\alpha + \beta \subset \alpha + \gamma$ is clear.

If $\alpha + \beta = \alpha + \gamma$, the cancellation law (Proposition 1.14) gives $\beta = \gamma$. Contradiction.

Thus $\alpha + \beta \subsetneq \alpha + \gamma$, i.e., $\alpha + \beta < \alpha + \gamma$.

Step 5: Consequence

Remark

It also follows that

$$\alpha > \mathsf{O}^* \quad \Longleftrightarrow \quad -\alpha < \mathsf{O}^*.$$

With Steps 4 and 5, the first requirement of an ordered field (Definition 1.17) is satisfied.

Steps 6–7: Multiplication

Step 6: Multiplication for Positive Cuts

Definition: Multiplication on $\mathbb{R}^+$

For $\alpha, \beta \in \mathbb{R}^+$ (i.e., $\alpha > 0^*$ and $\beta > 0^*$), define $\alpha\beta$ to be the set of all $p \in \mathbb{Q}$ such that

$$p \leq rs \quad \text{for some } r \in \alpha,\ s \in \beta,\ r > 0,\ s > 0.$$

Definition: The Unit Cut

$1^* = \{q \in \mathbb{Q} : q < 1\}$

Step 6: Axioms (M) and (D) on $\mathbb{R}^+$

Verification

The axioms (M) of Definition 1.12 hold in $\mathbb{R}^+$ with 1^* as the identity:

(M1) $\alpha\beta$ is a cut (closure)

(M2) $\alpha\beta = \beta\alpha$ (commutativity)

(M3) $(\alpha\beta)\gamma = \alpha(\beta\gamma)$ (associativity)

(M4) $\alpha \cdot 1^* = \alpha$ (identity)

(M5) Multiplicative inverses exist in $\mathbb{R}^+$

The distributive law (D) also holds in $\mathbb{R}^+$.

The proofs parallel the addition axioms from Step 4.

Step 7: Extending Multiplication to All of $\mathbb{R}$

Definition: Full Multiplication

Set $\alpha \cdot \mathrm{o}^* = \mathrm{o}^* \cdot \alpha = \mathrm{o}^*$ for all α, and:

$$\alpha\beta = \begin{cases} (-\alpha)(-\beta) & \text{if } \alpha < \mathrm{o}^*,\ \beta < \mathrm{o}^* \\ -[(-\alpha)\beta] & \text{if } \alpha < \mathrm{o}^*,\ \beta > \mathrm{o}^* \\ -[\alpha \cdot (-\beta)] & \text{if } \alpha > \mathrm{o}^*,\ \beta < \mathrm{o}^* \end{cases}$$

Step 7: Completing the Ordered Field

Verification

The axioms (M) extend from $\mathbb{R}^+$ to all of $\mathbb{R}$ using the identity $\gamma = -(-\gamma)$ (Proposition 1.14).

The **distributive law** $\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$ is verified by cases on the signs of α, β, $\beta + \gamma$.

Conclusion of Steps 1–7

$\mathbb{R}$ is an ordered field with the least-upper-bound property.

Steps 8–9: Embedding $\mathbb{Q}$ into $\mathbb{R}$

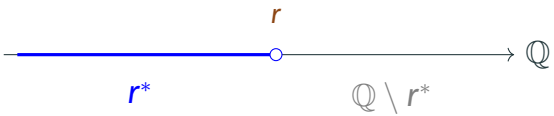
Step 8: Rational Cuts

Definition: Rational Cuts

For each $r \in \mathbb{Q}$, define

$$r^* = \{p \in \mathbb{Q} : p < r\}.$$

Each r^* is a cut (easy to verify). These are the rational cuts.



Step 8: Properties of Rational Cuts

Properties

For $r, s \in \mathbb{Q}$:

(a) $r^* + s^* = (r + s)^*$

Proof of (a)

($\subset$) If $p = u + v$ with $u < r$, $v < s$, then $p < r + s$. ($\supset$) If $p < r + s$, set $t = (r + s - p)/2$, then $r' = r - t \in r^*$, $s' = s - t \in s^*$, and $p = r' + s'$.

Step 8: Properties of Rational Cuts

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For $r, s \in \mathbb{Q}$:

(a) $r^* + s^* = (r + s)^*$

(b) $r^* \cdot s^* = (rs)^*$

The proof of (b) is similar to (a).

Step 8: Properties of Rational Cuts

Properties

For $r, s \in \mathbb{Q}$:

(a) $r^* + s^* = (r + s)^*$

(b) $r^* \cdot s^* = (rs)^*$

(c) $r^* < s^*$ if and only if $r < s$

Proof of (c)

If $r < s$: then $r \in s^*$ but $r \notin r^*$, so $r^* \subsetneq s^*$.

If $r^* < s^*$: $\exists\, p \in s^*$ with $p \notin r^*$, so $r \leq p < s$, giving $r < s$.

Step 9: $\mathbb{Q}$ as a Subfield of $\mathbb{R}$

Key Observation

The map $r \mapsto r^*$ from $\mathbb{Q}$ to $\mathbb{R}$ preserves:

- Sums: $r^* + s^* = (r + s)^*$
- Products: $r^* \cdot s^* = (rs)^*$
- Order: $r < s \iff r^* < s^*$

We say $\mathbb{Q}$ is isomorphic to $\mathbb{Q}^* = \{r^* : r \in \mathbb{Q}\}$.

This identification allows us to regard $\mathbb{Q}$ as a subfield of $\mathbb{R}$.

Step 9: Uniqueness and Historical Notes

Uniqueness

Any two ordered fields with the least-upper-bound property are isomorphic.

Thus Theorem 1.19 *characterizes* $\mathbb{R}$ completely.

Historical note:

- **Dedekind** (1872): Construction via cuts (the one we just saw)
- **Cantor** (1872): Construction via Cauchy sequences of rationals

Summary

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Key takeaways from this appendix:

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- A Dedekind cut is a nonempty, proper, downward-closed subset of $\mathbb{Q}$ with no largest element
- The set $\mathbb{R}$ of all cuts, ordered by inclusion, has the least-upper-bound property (the supremum is the union)
- With addition and multiplication defined on cuts, $\mathbb{R}$ becomes an ordered field
- The rational cuts $r^* = \{p \in \mathbb{Q} : p < r\}$ embed $\mathbb{Q}$ as a subfield of $\mathbb{R}$, completing the proof of Theorem 1.19

Coming up next: Chapter 2 — Basic Topology.